



Xi'an Jiaotong-Liverpool University

西交利物浦大學

FIN302

Finance and Markets

Lecture 10: Option Properties and Trading

Prof Tony He

Lecture 10- Option Properties and Trading

Questions and Learning Outcomes

- What are the options and how do they work?
 - Options: Types, Payoffs, and Profits
- What are the properties of options?
 - Put-Call Parity and Capital Structure
 - Early Exercise of American Options
 - The Effect of Dividends
- What are option trading strategies?
 - Naked and Covered Option Positions
 - Bull, Bear, and Butterfly Spreads



Lecture 10- Option Properties and Trading

- Readings:
 - Hull (2022), Chapters 10, 11, 12 (optional)
- Tutorial and Homework Problems from Hull:
 - 10.11-13, 18, 20, 22, 24
 - 11.9, 10, 12, 17, 18

Options: Types

- Option Types:
 - **Call**: gives the holder the *right to buy* the underlying asset at or by a certain date, for a certain price.
 - **Put**: gives the holder the *right to sell* the underlying asset at or by a certain date, for a certain price.

- Two possibilities for when the holder of an option may exercise his right to buy or sell the underlying asset:
 - **European** option: may be exercised *only when it matures*.
 - **American** option: may be exercised at *any time prior to its maturity*.

Options: Payoffs

➤ Notations:

- S_0 : the current price of the underlying stock
- T : the expiry date or maturity date.
- K : the exercise price or strike price.
- S_T : the price of the underlying asset at maturity

➤ Payoffs:

- **Long call** = $\max\{S_T - K, 0\}$
- **Short call** = $-\max\{S_T - K, 0\} = \min\{K - S_T, 0\}$
- **Long put** = $\max\{K - S_T, 0\}$
- **Short put** = $-\max\{K - S_T, 0\} = \min\{S_T - K, 0\}$

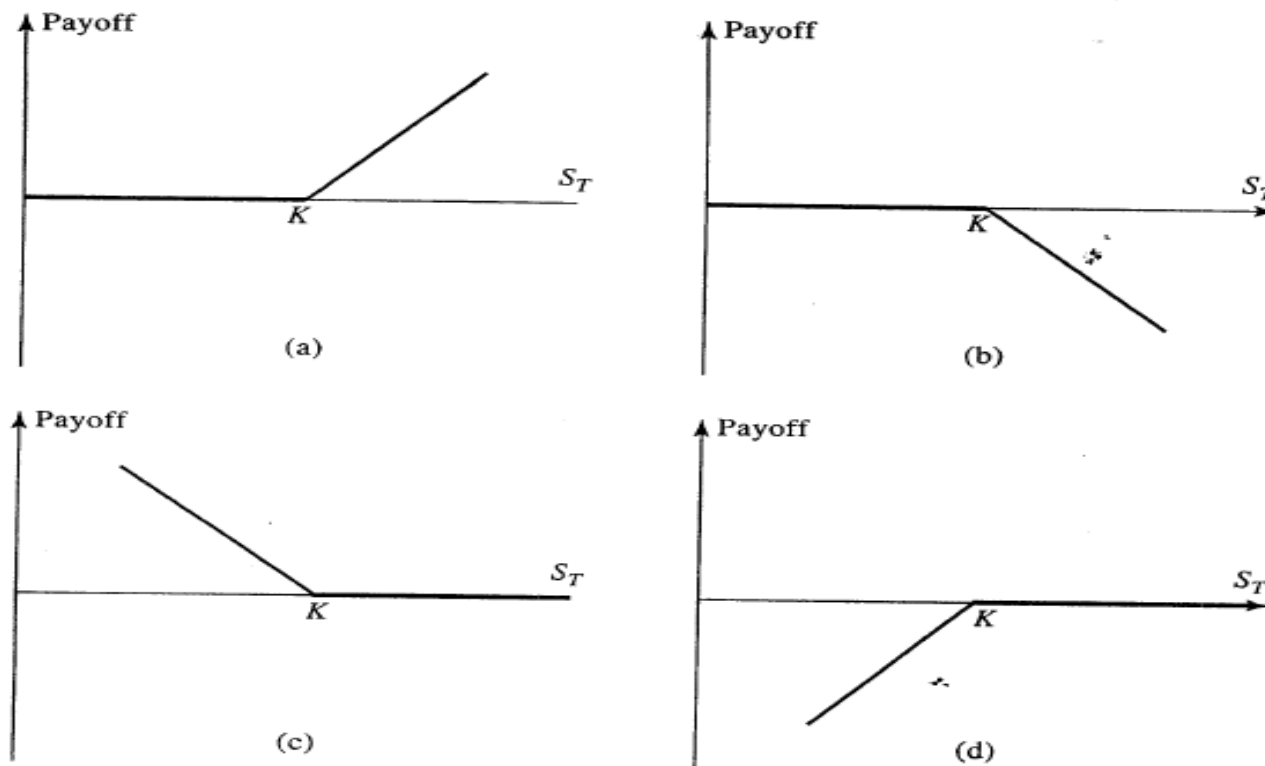
➤ Option Price (Premium):

- European: Call: c Put: p
- American: Call: C Put: P

- **Zero-sum game**: the payoffs of a short call and a short put are simply the negated payoffs of a long call and a long put, respectively.

Options: Payoffs

Figure 1: The payoffs of the four basic options positions: (a) a long call; (b) a short call; (c) a long put; and (d) a short put.



Option Moneyness

➤ Moneyness

- Call (put) is **in-the-money (ITM)** when $S_t > K$ ($S_t < K$);
- Call (put) is **out-the-money (OTM)** when $S_t < K$ ($S_t > K$);
- Call/put is **at-the-money (ATM)** when $S_t = K$

➤ The **value of an option** consists of its *intrinsic value* and its *time value*:

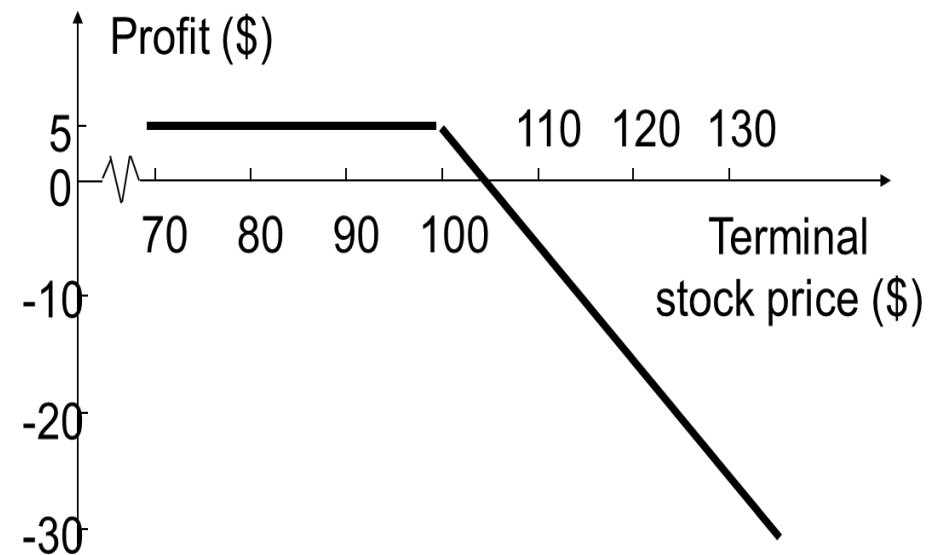
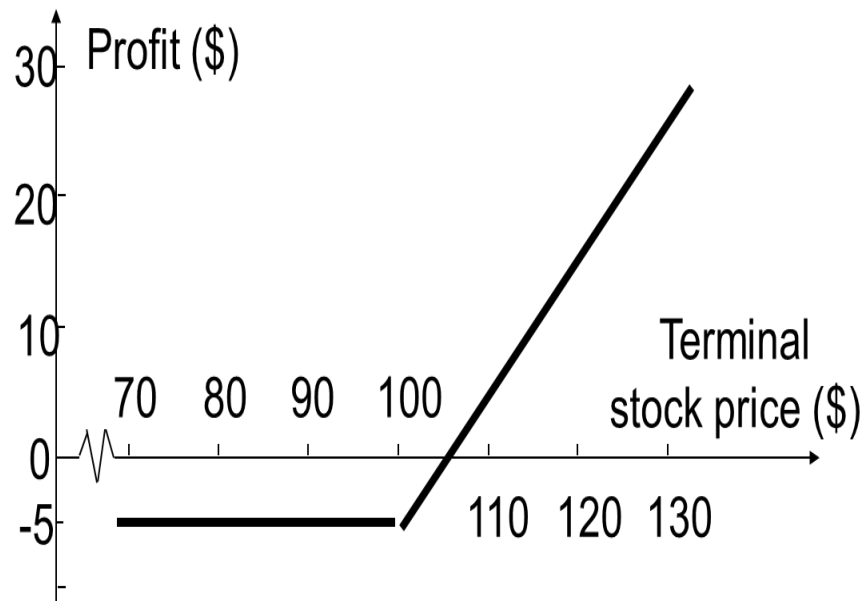
- The **intrinsic value** is the option's value if the option could be exercised immediately. ITM options have positive intrinsic value, while OTM options have no intrinsic value.
- The **time value** of the option becomes zero at the option's maturity.

Option Profits

- It is important to understand the **difference between the *payoff* of an option position and its *profit*.**
 - The payoff is the cash flow that occurs upon exercise.
 - The profit is the payoff plus the initial cash flow.
- There are two ways to understand either the payoff or the profit from an options strategy:
 - Construct a *table of payoffs* or a *table of profits*.
 - Sketch the *payoff diagram* or the *profit diagram*.

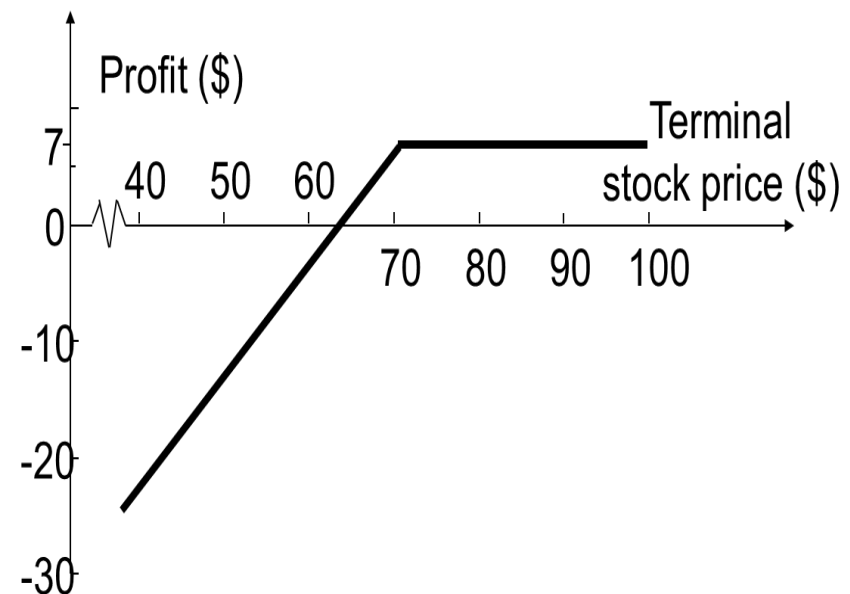
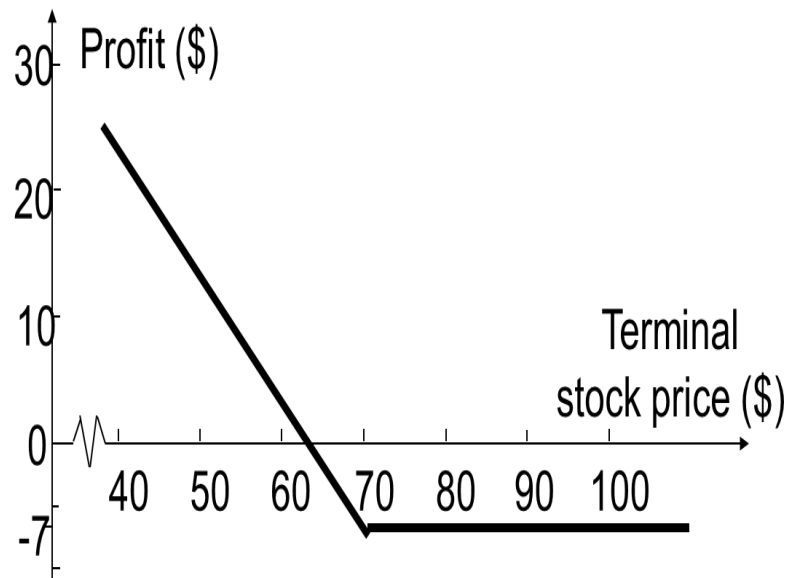
Option Profits: Calls

- **Long call** (left): Profit from **buying** one European **call** option: option price = \$5, strike price = \$100, option life = 2 months
- **Short call** (right): Profit from **writing (or selling)** one European **call** option: option price = \$5, strike price = \$100



Option Profits: Puts

- **Long put** (left): Profit from **buying** one European **put** option: option price = \$7, strike price = \$70
- **Short put** (right): Profit from **writing (or selling)** one European **put** option: option price = \$7, strike price = \$70



Option Price Properties

- **Six factors** that affect the price of a stock option:
 - S_0 : the current stock price
 - K : the strike price
 - T : the time to maturity
 - σ : the stock price volatility
 - r : the risk-free interest rate
 - D : the expected dividends during the life of the contract
- S_0 : Call price c increases and put price p decreases as the stock price S_0 increases
- K : Call price c decreases and put price p increases as the strike price K increases
- T : Call and put prices increases in T , the time to maturity, in general.



Option Price Properties

- The volatility σ : measures the uncertainty of future stock prices.
 - As σ increases,
 - the owner of a call (or put) benefits from the increased likelihood of extreme price increases (or decreases),
 - while the limited downside of call (or put) payoff protects him from the increased likelihood of extreme price decreases (or increases).
 - So the **option prices increase in σ** .

Option Price Properties

- The effect of **interest rates** on option prices is unclear.
 - As interest rate increases,
 - investors in the risky stock require a higher expected return, yet,
 - the present value of any future cash flow received by the owner of an option decreases.
 - The combined action of these **two effects** *increases the prices of call options* and *decreases the value of put options*.
- **Dividends** reduce the stock price on the ex-dividend date.
 - Call prices decrease in the sizes of anticipated future dividends.
 - Put prices increase in the sizes of anticipated future dividends.

Option Price Properties

- Table 1 summarizes the dependencies discussed so far.

<i>Variable</i>	<i>European call</i>	<i>European put</i>	<i>American call</i>	<i>American put</i>
Current stock price	+	–	+	–
Strike price	–	+	–	+
Time to expiration	?	?	+	+
Volatility	+	+	+	+
Risk-free rate	+	–	+	–
Amount of future dividends	–	+	–	+

+ indicates that an increase in the variable causes the option price to increase or stay the same;
– indicates that an increase in the variable causes the option price to decrease or stay the same;
? indicates that the relationship is uncertain.

Put-Call Parity for European Options

- Consider a portfolio with the following positions:
 - A **long** position in a non-dividend-paying **stock**
 - A **short** position in a European **call** on the stock
 - A **long** position in a European **put** on the stock
- The value of this portfolio is $S_0 - c + p$
- When its constituent options mature, the portfolio will produce the following payoff:

$$\begin{aligned} & S_T - \max\{S_T - K, 0\} + \max\{K - S_T, 0\} \\ &= \begin{cases} S_T - (S_T - K) + 0 & \text{if } S_T \geq K \\ S_T - 0 + (K - S_T) & \text{if } S_T < K \end{cases} \\ &= K. \end{aligned}$$



Put-Call Parity for European Options

- Another way of obtaining exactly the same payoff at time T is by investing Ke^{-rT}
- If the market is free of arbitrage opportunities we must have

$$S_0 - c + p = Ke^{-rT}$$

- or

$$c + Ke^{-rT} = p + S_0$$

- This identity is known as **put-call parity**.
- Any violation of put-call parity produces an arbitrage opportunity.

Put-Call Parity for European Options

- **Example:** Consider a European call strike at \$30 on a non-dividend-paying stock whose price is \$31.
- The option matures after three months, is trading for \$3 and the risk-free interest rate is 10% per annum, with continuous compounding.
- If a three-month European put on the stock, with the same strike, is trading for \$2.25, then put-call parity does not hold since

$$\begin{aligned} c + Ke^{-rT} &= 3 + 30 e^{-0.1 \times \frac{3}{12}} = 32.26 \\ &< 33.25 = 2.25 + 31 = p + S_0 \end{aligned}$$

Put-Call Parity for European Options

Example (cont.): An arbitrageur could do the following:

- Short-sell the stock at \$31
- Sell the put at \$2.25
- Purchase the call at \$3
- Invest $31 + 2.25 - 3 = 30.25$ at 10% for 3-month
- There are two possibilities when the options mature:
 - If $S_T \geq 30$, he exercises the call and the put expires worthless.
 - If $S_T < 30$, his counter-party exercises the put and the call expires worthless.
 - In either case he purchases the stock for \$30 to close-out his short position, yielding a total arbitrage profit of

$$30.25 e^{0.1 \frac{3}{12}} - 30 = 1.02$$



Capital Structure

- Company's asset A is financed by
 - **zero-coupon bonds** with face value of K at time T
 - **equity** with no-dividend payment
- At T
 - If $A_T \geq K$, equity holders pay K to bondholders
 - If $A_T < K$, equity holders are bankrupt and the company is owned by bond holders
 - **The value of the equity**: $\max(A_T - K, 0)$
 - **The value of the bond**: $\min(A_T, K) = K - \max(K - A_T, 0)$

Capital Structure

➤ The value at time 0:

- Equity: c
- Bond: $PV(K) - p$
- Company: $A_0 = c + [PV(K) - p]$
- Put-call parity:
$$c + PV(K) = p + A_0$$

Early Exercise Premium

- Since the owner of an American option may always choose to hold it to maturity, American options are always at least as valuable as the equivalent European contracts:

$$C \geq c, \quad P \geq p$$

- The differences $C - c$ and $P - p$ are referred to as the **early exercise premiums** of American calls and puts, respectively



Put-Call Parity for American Options

- Put-call parity does not apply to American options.
- However, American options on a non-dividend-paying stock do satisfy the following put-call parity-like inequalities:

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$

- **Example:** Consider a five-month American call strike at \$20 on a non-dividend-paying stock whose price is \$19. If the price of this option is \$1.50 and the continuously compounded risk-free interest rate is 10% per annum, then the price of a similarly strike five-month American put satisfies

$$S_0 - K \leq C - P \leq S_0 - Ke^{-rT}$$
$$C + Ke^{-rT} - S_0 \leq P \leq C + K - S_0$$

$$1.5 + 20e^{-0.1 \frac{5}{12}} - 19 \leq P \leq 1.5 + 20 - 19$$
$$1.68 \leq P \leq 2.5$$



Early Exercise of American Call

- Using the lower bound for the price of an American call on a non-dividend-paying stock, we obtain

$$C \geq \max\{0, S_0 - Ke^{-rT}\} > \max\{0, S_0 - K\}$$

- Thus the price of an American call on a non-dividend-paying stock is always strictly greater than the payoff received by the holder in the event of early exercise.
- **Early exercise of an American call on a non-dividend-paying stock is never optimal.** Therefore $C = c$.
 - **Time value of money**---delay the payment if call is in-the-money
 - **Insurance**---pay no more than K ; pay less if call is out-of-the-money.



Early Exercise of American Put

- Suppose the current price of a non-dividend-stock satisfies
$$S_0 < K(1 - e^{-rT})$$
- Then the payoff from immediately exercising an American put is
$$K - S_0 > K - K(1 - e^{-rT}) = Ke^{-rT} \geq e^{-rT} \max\{K - S_T, 0\}$$
 - *Immediate early exercise of an American put on a non-dividend-paying stock is therefore optimal if the stock price is low enough.* Therefore $P > p$
- **Two reasons:**
 - *Time value of money:* If the price of a non-dividend-paying stock is sufficiently low, then the payoff from immediately exercising an American put exceeds the present-value of the put payoff at maturity, irrespective of the stock price at that time.
 - *Insurance:* to sell for at least K

The Effect of the Dividends

- Consider a dividend-paying stock, and let D denote the present-value of all dividends paid during the life of an option
- In that case the lower bounds for European stock options are:
$$c \geq S_0 - D - Ke^{-rT}, \quad p \geq Ke^{-rT} + D - S_0$$
- For an **American call** on a dividend-paying stock:
 - **Early exercise may be optimal** immediately prior to an ex-dividend date
 - Early exercise is not optimal at any other time
- The **put-call parity for European options on a dividend-paying stock** is

$$c + D + Ke^{-rT} = p + S_0$$

- For American options on a dividend-paying stock we have

$$S_0 - D - K \leq C - P \leq S_0 - Ke^{-rT}$$



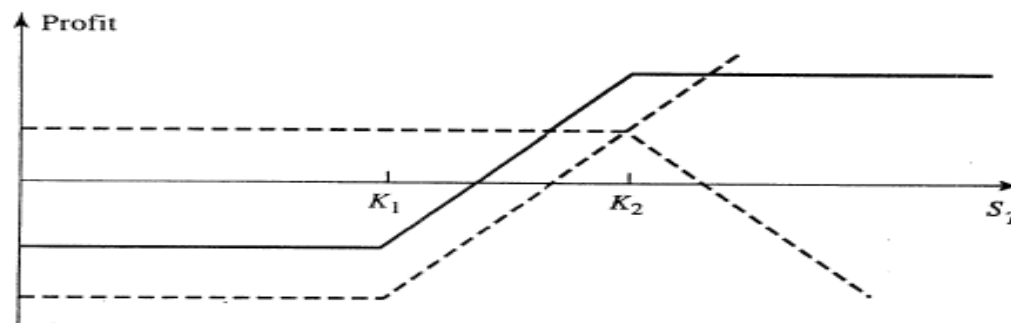
Option Trading Strategies

- Four basic ***naked option*** positions:
 - long call, long put, short call, short put
- **Covered options:**
 - ***Covered call***: long asset + short call
 - To protect short call when call is ITM
 - Equivalent to a short put: put-call parity
 - ***Covered (protective) put***: long asset + long put.
 - To protect asset value when put is ITM
 - Equivalent to a long call: put-call parity

Option Trading Strategies: Bull Spread

- A **spread** is an options strategy involving two or more options of the same type.
- A **bull spread**: purchasing a call at K_1 and selling a call at K_2 , with the same maturity and $K_1 < K_2$ to *profit from bull market* and *limit the loss from bear market*.
- **Example**: Constructs a bull spread by purchasing a \$30 call for \$3 and selling a \$35 call for \$1.

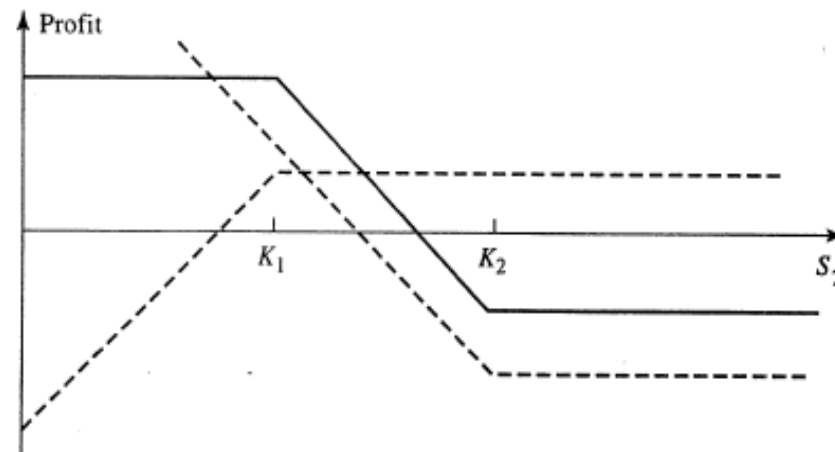
Stock price range	Long Call (K=30)	Short call (K=35)	Premium	Profit/loss
$S_T \leq 30$	0	0	$-3 + 1 = -2$	-2
$30 \leq S_T < 35$	$S_T - 30$	0	-2	$S_T - 32$
$S_T \geq 35$	$S_T - 30$	$35 - S_T$	-2	3



Option Trading Strategies: Bear Spread

- A **bear spread**: selling a put at K_1 and purchasing a put at K_2 , with the same maturity and $K_1 < K_2$ to *profit from bear market* and *limit the loss from bull market*.
- **Example**: An investor constructs a bear spread by selling a \$30 put for \$1 and purchasing a \$35 put for \$3.

Stock price range	Long Put (K=35)	Short Put (K=30)	Premium	Profit/loss
$S_T \leq 30$	$35 - S_T$	$S_T - 30$	$-3 + 1 = -2$	3
$30 \leq S_T < 35$	$35 - S_T$	0	-2	$33 - S_T$
$S_T \geq 35$	0	0	-2	-2



Option Trading Strategies: Butterfly Spread

- A **butterfly spread**: purchasing a call at strike K_1 and a call at strike K_3 , and selling two calls at strike K_2 , with the same maturity and

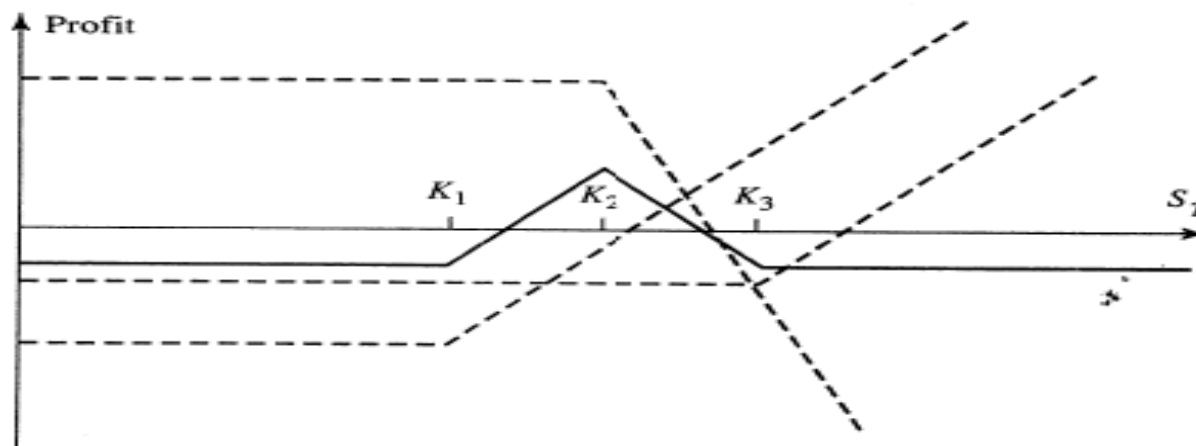
$$K_1 < K_2 < K_3 \quad \text{and} \quad K_3 - K_2 = K_2 - K_1$$

- In general, K_2 is close to the prevailing price of the underlying asset when the butterfly spread is constructed,
- It profits when the price of the underlying asset will not be volatile.

Option Trading Strategies: Butterfly Spread

Example: The market prices of three six-months calls on a certain stock that is currently trading for \$61 per share are $c_1 = \$10$ for $K_1 = \$55$, $c_2 = \$7$ for $K_2 = \$60$, and $c_3 = \$5$ for $K_3 = \$65$. An investor who believes the stock price will not be volatile over the next six months uses these options to construct a butterfly spread. The initial cost of the strategy to the investor is $10 - 2 \times 7 + 5 = \$1$.

Stock price range range	Long Call (K=55) (K=55)	2 Short Calls (K=60) (K=60)	Long Call (K=65) (K=65)	Premium	Profit/loss
$S_T \leq 55$	0	0	0	-1	-1
$55 \leq S_T < 60$	$S_T - 55$	0	0	-1	$S_T - 56$
$60 \leq S_T < 65$	$S_T - 55$	$-2(S_T - 60)$	0	-1	$64 - S_T$
$S_T \geq 65$	$S_T - 55$	$-2(S_T - 60)$	$S_T - 65$	-1	-1



Appendix: Additional Reading

Option Price Bounds

➤ Upper Bounds:

- European and American call:

$$c \leq S_0, \quad C \leq S_0$$

- European and American put:

$$p \leq Ke^{-rT}, \quad P \leq K$$

➤ Lower Bounds:

- The European call on a non-dividend-paying stock must satisfy the following two lower bounds:

$$c \geq 0, \quad c \geq S_0 - Ke^{-rT} \rightarrow c \geq \max\{0, S_0 - Ke^{-rT}\}$$

- Since $C \geq c$,

$$C \geq \max\{0, S_0 - Ke^{-rT}\}$$



Option Price Bounds

- To justify the second bound, $c \geq S_0 - Ke^{-rT}$, suppose $c < S_0 - Ke^{-rT}$, then an arbitrageur could do the following:
 - Short-sell the stock
 - Purchase the call
 - Invest $S_0 - c > Ke^{-rT}$ in a risk-free money-market account.
- There are two possibilities when the option matures:
 - If $S_T \geq K$, he purchases the stock for K to close out his short position, for a total profit of
$$(S_0 - c)e^{rT} - K > Ke^{-rT}e^{rT} - K = 0$$
 - If $S_T < K$, he purchases the stock for S_T in the spot market to close out his short position, for a total profit of
$$(S_0 - c)e^{rT} - S_T > Ke^{-rT}e^{rT} - K = K - K = 0$$
 - Resulting in an arbitrage profit,

Option Price Bounds

➤ Lower Bounds:

- The price of a European put on a non-dividend-paying stock must satisfy the following two lower bounds:

$$p \geq 0, \quad p \geq Ke^{-rT} - S_0 \rightarrow \mathbf{p \geq \max\{0, Ke^{-rT} - S_0\}}$$

- Similar arbitrage argument can be used to verify this
- Since an American put on a non-dividend-paying stock can be exercised any time before the maturity, we have

$$\mathbf{P \geq \max\{0, K - S_0\}}$$

Option Price Bounds

Example: Consider an European put with strike at \$40 on a non-dividend-paying stock whose price is \$38. If the option matures after three months and the risk-free interest rate is 10% per annum, with continuous compounding, then the minimum price for the option is

$$p \leq Ke^{-rT} = 40e^{-0.1\frac{3}{12}} = 39.01$$
$$p \geq \max\{0, Ke^{-rT} - S_0\} \geq \max\left\{0, 40e^{-0.1\frac{3}{12}} - 38\right\} = 1.01$$

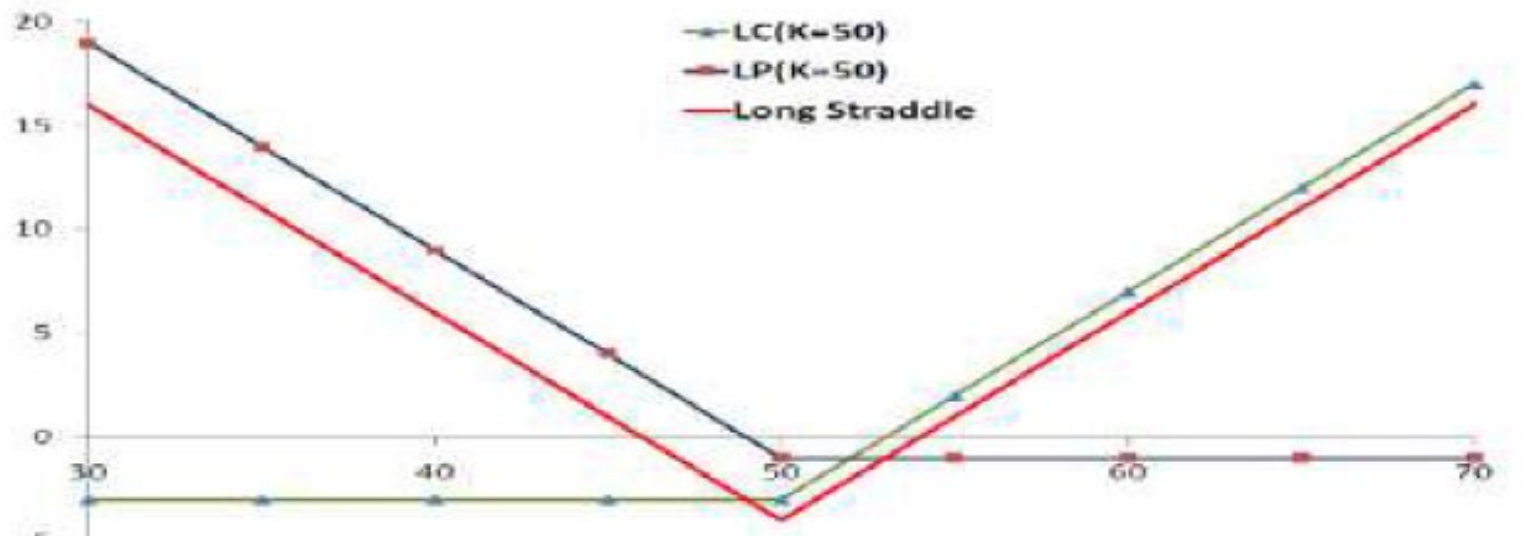
For the American option,

$$P \geq \max\{0, K - S_0\} = 2$$



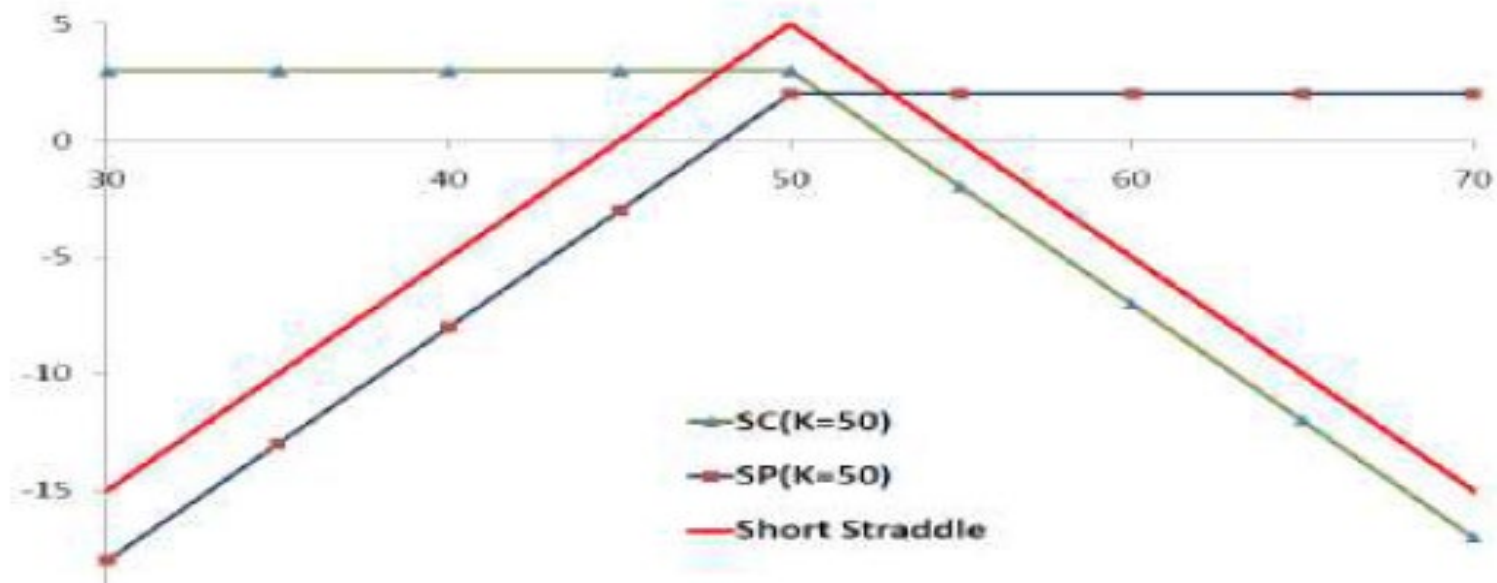
Option Trading: Combination---Straddle

- A **combination** is an options strategy that involves both puts and calls on the same underlying asset.
- A **straddle** (or bottom straddle): purchasing a put and a call with the same strike and the same maturity.
 - To profit when the price of the underlying asset is volatile.
 - Limited loss potential (options premium) in stable markets.



Option Trading Strategies: Combination---Straddle

- A **top straddle**: selling a put and a call with the same strike and the same maturity.
- To profit when the asset price is close to the strike, not volatile, but the potential for loss is unlimited, an extremely risky strategy taken by institutional traders.

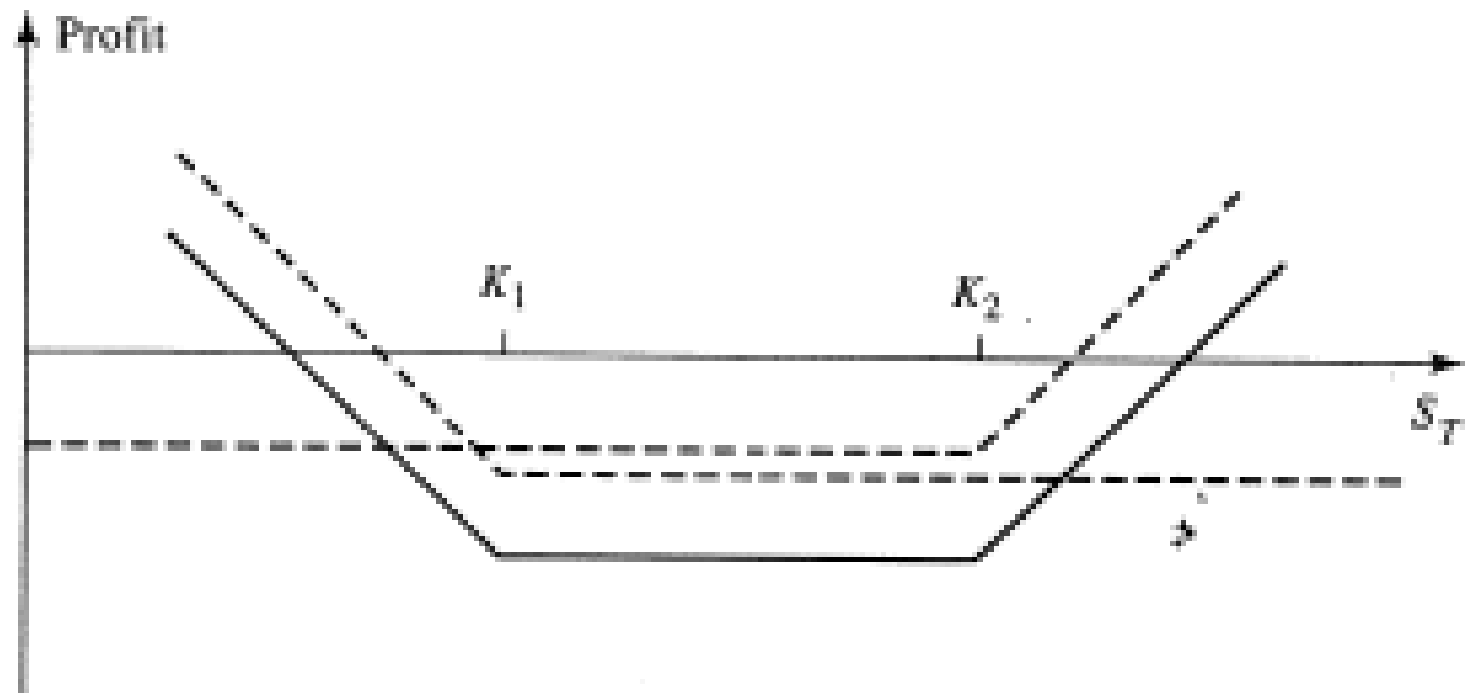


Option Trading Strategies: Strip & Strap

- Strips and straps are variations of a straddle.
 - A **strip**: purchasing two puts and one call with the same strike and the same maturity, to profit from a volatile and bearish market.
 - A **strap**: purchasing one put and two calls with the same strike and the same maturity, to profit from a volatile and bullish market.

Option Trading Strategies: Strangle

A **strangle** (or a *bottom vertical combination*): purchasing a put at strike K_1 and a call at strike K_2 , with the same maturity and $K_1 < K_2$, to profit from high volatility in either direction.



Option Trading: Market Makers and Margin

➤ Market Makers:

- To facilitate options trading
- Quotes both bid and ask prices when requested, without knowing whether the individual requesting the quotes wants to buy or sell

➤ Margin:

- When a naked option is written the margin is the greater of:
 - A total of 100% of the proceeds of the sale plus 20% of the underlying share price less the amount (if any) by which the option is out of the money
 - A total of 100% of the proceeds of the sale plus 10% of the underlying share price (call) or exercise price (put)

Employee Stock Options and Convertible Bonds

➤ Employee stock options

- a form of remuneration issued by a company to its executives
- usually at the money when issued
- when options are exercised the company issues more stock and sells it to the option holder

➤ Convertible bonds:

- regular bonds that can be exchanged for equity at certain times in the future according to a predetermined exchange ratio
- a convertible is callable, a way in which the issuer can force conversion at a time earlier than the holder might otherwise choose